

Rover Mission Requirements Audit

Turn a broad rover mission statement into clear, testable engineering requirements.

Intermediate	1 class period	Practice & Review
--------------	----------------	-------------------

Engineering Context

A client has requested “a reliable rover that can transport a payload across rough terrain.” The statement is not specific enough to design or verify.

What You Will Practice

- Distinguish needs from requirements
- Write measurable shall-statements
- Identify verification methods
- Detect conflicting or missing requirements

What You Need

- Mission statement
- Requirements template
- Engineering notebook
- Optional sample rover data

Student Instructions

- 1 Underline every user need or desired function in the mission statement.
- 2 Convert each need into one or more measurable requirements.
- 3 Add units, limits, conditions, and success criteria.
- 4 Assign a verification method: inspection, analysis, demonstration, or test.
- 5 Identify at least two missing requirements and one possible conflict.
- 6 Prioritize requirements as Must, Should, or Could.
- 7 Conduct a peer audit and revise unclear statements.

Engineering Requirements

- At least ten measurable requirements
- Every requirement uses a defined condition and limit
- Each requirement has a verification method
- Avoid solution-specific wording unless required by the client

Constraints & Safety

- Do not invent unrealistic performance limits
- Maintain traceability to the original mission need
- Clearly mark assumptions

What You Submit

- Need-to-requirement table
- Prioritized requirement list
- Verification matrix
- Conflict/missing-information note
- Revised final set

Reflection Questions

- 1 Which user need was hardest to make measurable?
- 2 What requirement could unintentionally restrict design creativity?
- 3 Which requirement has the highest mission risk?
- 4 What question should be returned to the client?

Engineering Tip A strong requirement states what the system must accomplish, not how the designer must accomplish it.	Common Mistake Words such as reliable, fast, lightweight, and easy are not testable without a number or condition.
Stretch Goal Add interface requirements for charging, payload attachment, or data transfer.	Career Connection Systems engineers manage requirements and verification across complex aerospace projects.